

CS 6120: Analyzing Bias Transfer In Model Distillation

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Abstract

This paper investigates whether bias present in large language models transfers to smaller student models through knowledge distillation. Using LLaMA 3.1 8B Instruct as the teacher, baseline and distilled versions of three student architectures (BoW, CNN, and DistilBERT) were trained on the IMDB reviews dataset for sentiment classification. Evaluation on the Equity Evaluation Corpus reveals systematic shifts in sentiment scores for gendered and race-associated sentences, demonstrating that social biases in the teacher are preserved and in some cases amplified during distillation. Among student models, distilled DistilBERT exhibits the largest and most consistent bias shifts, distilled BoW shows relatively small effects, and CNN falls in between. These findings indicate that model compression cannot be assumed to mitigate bias; higher-capacity students appear particularly prone to inheriting the teacher’s demographic asymmetries, though even less sophisticated architectures acquire measurable bias from the teacher.

1 Introduction

Large Language Models, LLMs, have become ubiquitous since ChatGPT’s release in November 2022, appearing in applications from search engines to e-commerce. Understanding the biases these models contain is critical given their widespread deployment. Additionally, the push to run models locally on smartphones has made model distillation an essential technique for creating compact, efficient models.

Model distillation (Hinton et al., 2015) trains a smaller student model to replicate a larger teacher model’s outputs, producing significantly smaller models with comparable performance on specific tasks, as demonstrated by DistilBERT (Sanh et al., 2020). Since teacher models are known to exhibit biases (Hu et al., 2024), a natural question arises: do distilled models inherit these biases?

This paper investigates whether and how biases transfer from LLMs to smaller models during distillation. Specifically, the study examines whether different student architectures learn biases to different extents. Sentiment analysis serves as the evaluation task because it enables straightforward binary classification while providing insight into a model’s implicit value associations with demographic groups.

2 Background

Hinton et al. (2015), introduced the idea of knowledge distillation, where smaller student models learn from the soft outputs of larger ones. This forms the foundation for which brought model distillation into fruition. Sanh et al. (2020), later demonstrated that transformers could be effectively distilled, compressing BERT into a smaller version with only a small drop in accuracy. Hu et al. (2024), demonstrated that most large language models show human-like group biases, but fine-tuning and cleaner data can reduce them. It was found that model compression methods like pruning and distillation affect the fairness of models (Ramesh et al., 2023). Ramesh et al. (2023) helped frame our investigation into bias and compression. Mohanty et al. (2023), found what kinds of information are lost during distillation. Their work gave additional motivation to study if bias is something that gets carried over or filtered out. Kiritchenko and Mohammad (2018) introduced the Equity Evaluation Corpus, which enables precise gender and race bias measurements through controlled sentence pairs. This corpus and its evaluation methodology form the foundation of the analysis presented in this paper.

3 Data

The IMDB Movie reviews dataset (Maas et al., 2011) was the main dataset used for the distillation, as it was already label allowing for a baseline

model to be trained and be used as a comparison to the distilled version. The IMDB Movie reviews dataset has 50,000 user written movie reviews labeled as positive or negative (split evenly), though a few duplicates in the training and testing datasets were found and therefore was fixed by combining the train and test dataset, then removing the duplicates, and then doing an even splitting into train and test (Refer to 1 for the exact split of the data). The text in this dataset is fairly noisy and contains slangs, weird punctuations, HTML tags and varied writing styles. Basic preprocessing was applied such as lower-casing all words, removing extra spaces, and tokenizing the text into individual words. The raw wording were mostly maintained as sentiment can depend on phrasing, but unknown or rare words were mapped to the "UNK" token the models can process them consistently.

To study the bias transfer, the Equity Evaluation Corpus (EEC) dataset is used which contains 8640 carefully constructed sentences (Kiritchenko and Mohammad, 2018). Unlike the IMDB movie reviews dataset, EEC is a controlled dataset made up of sentence templates¹ and varies only in demographic terms (such as race and or gender). This allows for detecting changes in a sensitive word sentiment prediction from the models, which is a direct way to measure bias. EEC requires no cleaning since the sentences are short and standardized. Tokenization is then performed the same as mentioned before.

4 Methods

To obtain the label probabilities from the teacher model, Llama 3.1 8B Instruct is prompted with a structured instruction format where the next token represents the sentiment label. The model outputs logits over the vocabulary; the logits corresponding to the "Negative" and "Positive" label tokens are extracted and passed through a softmax to obtain a probability distribution over the two labels. The same prompting scheme is used for both IMDB and EEC inputs (Figure 3).

Using the IMDB dataset and the teacher outputs, two versions of each student architecture are trained: a *baseline* model that uses the ground-truth IMDB labels as targets, and a *distilled* model that uses the teacher’s label probabilities as soft targets. Because the data, vocabulary, and training splits

are matched, differences between baseline and distilled variants can be attributed to the distillation process rather than changes in preprocessing.

To study how different architectures inherit bias from a large teacher, three student models are considered: a Neural Bag-of-Words (BoW) classifier, a Convolutional Neural Network (TextCNN), and DistilBERT. BoW and TextCNN reuse the same token-to-index mapping constructed for the IMDB baseline models, ensuring consistent input representations across conditions.

Neural Bag-of-Words (BoW) The BoW model uses a trainable embedding matrix followed by mean pooling over token embeddings to form a sentence representation, which is then passed through a linear classifier to produce logits for the two sentiment classes. This low-capacity architecture serves as a simple test of whether minimal models still inherit the teacher’s biases.

TextCNN The TextCNN model follows Kim (2014), adapted to the IMDB vocabulary. Inputs are mapped to embeddings, then processed by multiple one-dimensional convolutional filters with kernel sizes 4, 5, and 6. Max-over-time pooling is applied to each filter’s output, the pooled features are concatenated, and a linear layer produces sentiment logits. Compared to BoW, the CNN captures limited word-order information while remaining relatively lightweight.

DistilBERT DistilBERT (Sanh et al., 2020), a compact transformer encoder distilled from BERT, is used as a higher-capacity student. The same DistilBERT backbone is fine-tuned for sentiment classification; the distilled variant receives supervision from the teacher’s soft label distributions. Training follows the standard DistilBERT distillation setup described in Sanh et al. (2020).

Bias Evaluation Methodology Bias transfer from teacher to students is quantified using the Equity Evaluation Corpus (EEC) and the evaluation framework of Kiritchenko and Mohammad (2018). EEC consists of controlled sentence templates in which a single demographic term (gendered word or race-associated name) is varied while all other content is held fixed, so any difference in predicted sentiment can be attributed to that term.

For gender, matched pairs such as “my daughter feels happy” vs. “my son feels happy” are considered. Let s_f and s_m denote the positive sentiment

¹Refer to Table 1 in Kiritchenko and Mohammad, 2018 for the 11 sentence templates.

scores for the female and male variants; the gender difference for each pair is $\Delta_{\text{gender}} = \bar{s}_f - \bar{s}_m$. For race, EEC templates are instantiated with lists of names associated with African-American (AA) or European-American (EA) identity. For each template–emotion combination, mean scores for AA and EA names are computed and subtracted to obtain $\Delta_{\text{race}} = \bar{s}_{AA} - \bar{s}_{EA}$.

Following Kiritchenko and Mohammad (2018), the sets of Δ_{gender} and Δ_{race} values are treated as paired observations and evaluated with two-sided paired t -tests against a zero-mean null hypothesis. Bonferroni correction is applied over all model–demographic comparisons to control the family-wise error rate. For each model, the analysis reports mean bias differences and Bonferroni-corrected p -values, enabling direct comparison of baseline and distilled variants and quantifying the extent to which bias is transferred from the teacher.

5 Results

All models achieved F1 scores above 85%, indicating strong performance on binary sentiment classification (Refer to table 2). Given this accuracy, evaluation on the EEC dataset provides meaningful insight into bias transfer. However, Figure 4 reveals that student models produced inconclusive predictions across emotion categories, whereas Llama consistently assigned sentiment aligned with each sentence’s emotional content. This suggests that EEC lies outside the distribution of the IMDB training data. While this distribution mismatch is important to note, meaningful results can still be extracted: each distilled model has a corresponding baseline trained on identical data, so differences between them isolate the effect of distillation and allow bias transfer to be understood independently of absolute prediction quality. Figures 1 and 2

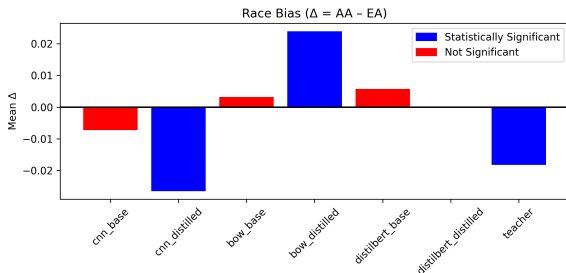


Figure 1: **Race Bias across all teacher and student models. Positive Δ indicates higher sentiment for African-American names.**

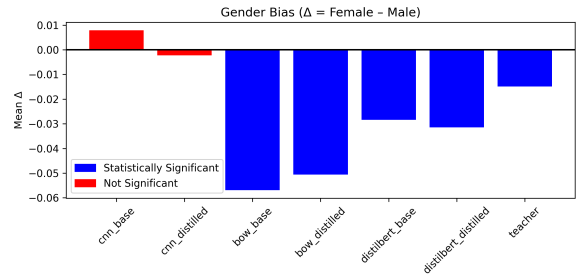


Figure 2: **Gender Bias across all teacher and student models. Positive Δ indicates higher sentiment for female terms.**

present the aggregate race and gender bias measurements across all teacher and student models. Each bar represents the mean sentiment difference between paired sentences differing only in a demographic attribute, with blue bars indicating statistical significance after Bonferroni correction.

Across both gender and race categories, the teacher model (Llama) exhibits measurable bias, which aligns with findings from prior works (Kiritchenko and Mohammad, 2018; Hu et al., 2024). Interestingly, several distilled student models display *stronger* demographic biases than their baseline counterparts, despite having similar task accuracy. This suggests that bias can intensify during distillation, even when the student model successfully learns the teacher’s task-level behavior.

Another high-level trend is that gender bias appears consistently larger in magnitude than race bias across all models. In particular, multiple student models show significantly more negative sentiment toward male-associated sentences, whereas race effects tend to be smaller and more model-dependent. These overall patterns motivate a deeper model-by-model analysis, presented next.

Bag of Words Distilled Model: The distilled BoW model remains very close to its baseline on EEC: a large fraction of sentences change only slightly in predicted sentiment, and race effects are small. However, the distilled variant shifts male-gendered sentences somewhat more negative than female ones, with sadness-related templates showing the largest differences. These patterns indicate that even a low-capacity BoW student inherits mild gender bias from the teacher while largely ignoring race-related asymmetries (see Appendix C.3 for detailed distributions).

Convolutional Neural Network Model: The distilled CNN exhibits much larger changes than

BoW: almost all EEC sentences receive different sentiment scores, and many shifts are substantial. Gender effects become more polarized, with female terms often receiving more negative sentiment than in the baseline, and race effects strengthen as well, generally favoring European-American over African-American names. Overall, the CNN’s greater capacity appears to absorb and amplify the teacher’s latent demographic biases more aggressively than the BoW student (see Appendix C.4).

DistilBERT Model: DistilBERT is the most expressive student and achieves the highest IMDB performance, yet its distilled variant still shows notable EEC shifts. Overall, sentiment changes are larger than for BoW but less volatile than for the CNN, and are relatively evenly distributed across demographic groups. Gender differences are clearly present, whereas race differences are small and not statistically significant after correction, suggesting that distillation does not strongly transmit Llama’s race bias in this setting. The combination of strong task accuracy and systematic gender effects underscores that high performance under distillation does not guarantee fairness (full breakdown in Appendix C.5).

6 Conclusions

This study demonstrates that bias present in a large teacher model is transferred to smaller student models through distillation. Using Llama 3.1 8B as the teacher and three student architectures (BoW, CNN, and DistilBERT), sentiment predictions were distilled on IMDB and demographic effects were then measured on the Equity Evaluation Corpus. Across models, systematic shifts in sentiment scores for gendered and race-associated sentences were observed, indicating that social biases in the teacher do not vanish in the distillation process. Instead, they are often preserved and, in some cases, amplified. Among the student models, the distilled DistilBERT model exhibited the largest and most consistent bias shifts, the distilled BoW model showed relatively small effects, and the CNN fell in between (Figures 1 and 2).

These results highlight an important implication: model compression is not inherently a bias mitigation strategy. Even when student models match or approach the teacher’s task performance, their demographic behavior can differ in non-trivial ways. Higher-capacity students, such as DistilBERT, may inherit more of the teacher’s

nuanced preferences, including undesirable asymmetries across demographic groups. Practitioners deploying distilled models—particularly in edge or resource-constrained settings—should therefore evaluate not only accuracy but also fairness characteristics after distillation.

Several limitations constrain the generality of these findings. The IMDB dataset is narrow in domain and stylistically distinct from EEC, producing the wide variation in predicted sentiment seen in Figure 4; a more diverse, conversational corpus would yield more representative behavior. Demographic coverage is also limited to two genders and a small set of African-American and European-American names—extending the analysis to non-binary identities and a broader range of racial and ethnic groups would strengthen the conclusions. Finally, only a single teacher and three student architectures, each trained once, are examined. The sensitivity of bias transfer to teacher choice (e.g., aligned vs. uncensored models), model size, or random initialization remains unclear and warrants further study.

An important direction for future work is studying designs for distillation procedures that explicitly mitigate bias rather than treating it as a side effect, such as incorporating fairness-aware loss terms or constraining predictions on controlled datasets like EEC. More broadly, any pipeline using LLMs as teachers should treat fairness evaluation as a primary objective: distillation can make models cheaper and faster, but without deliberate intervention it can also quietly propagate the teacher’s social biases.

Acknowledgments

While all the writing was originally manually written, AI was used for revisions during the writing process (ChatGPT 5.1 and Claude Opus 4.5). Any AI used has been read for correctness and revised to correct any mistakes it made. Any uses of AI in the code have been documented as comments.

Code

The code and all the datasets for the distillation and analysis can be found at github.com/CharlesC03/NLP_Final_Project.

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A IMDB Dataset Split

Split	Positive	Negative	Total
Train	12,433	12,358	24,791
Test	12,451	12,340	24,791
Total	24,884	24,698	49,582

Table 1: IMDB dataset split composition.

B Teacher Model Prompt

Classify the sentiment of the
 ↪ following texts as either Negative
 ↪ , or Positive.

Text: This is such a crappy movie
 ↪ I have no idea how it got on
 ↪ the shelves, they must have
 ↪ paid the movie store to make
 ↪ them put it there, seriously
 ↪ !...
 Sentiment: Negative

Text: Wow! I have seen so many
 ↪ bad low budget films lately,
 ↪ but this one is great. ...
 Sentiment: Positive

Text: I’ve seen tons of science
 ↪ fiction from the 70s; some
 ↪ horrendously bad, and others
 ↪ thought provoking and truly
 ↪ frightening. Soylent Green
 ↪ fits into the latter category.
 ↪ ...
 Sentiment: Positive

Text: [FILL IN HERE]
 Sentiment:

Figure 3: Example of Prompt Getting Text Sentiment.

C Model Performance

C.1 IMDB Test Accuracy

Model	Accuracy	Precision	Recall	F1 Score
Llama 3.1 8B Instruct	0.9458	0.9242	0.9717	0.9474
BoW Baseline	0.8765	0.8577	0.9042	0.8803
BoW Distilled	0.8693	0.8395	0.9147	0.8755
CNN Baseline	0.8703	0.8738	0.8669	0.8703
CNN Distilled	0.8702	0.8651	0.8786	0.8718
DistilBERT Baseline	0.9662	0.9649	0.9678	0.9664
DistilBERT Distilled	0.9180	0.8898	0.9551	0.9213

Table 2: Model Performance on the IMDB Test Dataset.

C.2 Overall Emotional Prediction Graphs

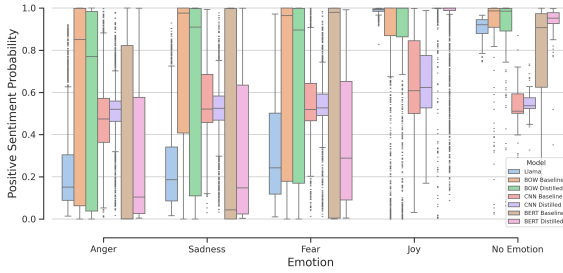


Figure 4: **Positive Sentiment Probability for each Model by Emotion Contained in the Sentence:** This shows the distribution of the probability a sentence with a given emotion would be positive given the model.

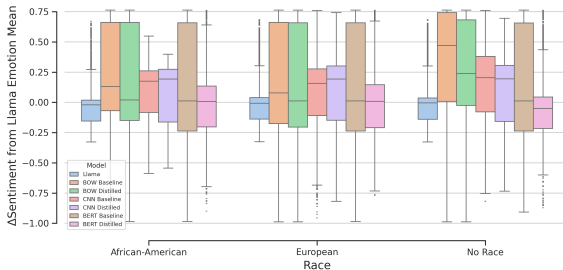


Figure 5: **Distribution of Model Deviations from Llama's Mean Positive Sentiment Predictions by Race.** Box plots showing each model's deviation from Llama's mean positive sentiment prediction per emotion to account for each model's varying prediction trends across emotion categories. The mean is performed from each emotion as it provides a more meaningful comparison between models by controlling for emotion-specific baselines. Results are grouped by race, offering a more detailed view of Figure 1 by illustrating the full distribution of prediction differences.

Llama was the only model confident in its predictions (Refer to Figure 4), indicating that IMDB was not representative of the EEC dataset distribution. This distribution mismatch was especially pronounced for the baseline DistilBERT model, though performance improved in the distilled version. This suggests that during distillation, DistilBERT was able to generalize better than its baseline counterpart. Figures 5 and 6 further illustrate the demographic biases the student models acquired. Notably, all student models shifted in the direction of the teacher model across demographic groups. Additionally, while the majority of predictions showed little change, the largest shifts occurred among outliers. Therefore, distillation may improve model generalization even when training

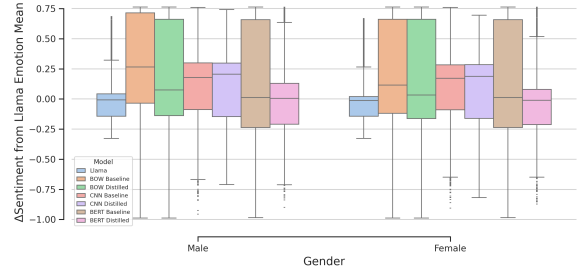


Figure 6: **Distribution of Model Deviations from Llama's Mean Positive Sentiment Predictions by Gender.** Box plots showing each model's deviation from Llama's mean positive sentiment prediction per emotion to account for each model's varying prediction trends across emotion categories. The mean is performed from each emotion as it provides a more meaningful comparison between models by controlling for emotion-specific baselines. Results are grouped by gender, offering a more detailed view of Figure 2 by illustrating the full distribution of prediction differences.

data is not representative of the target task, with higher-capacity models better imitating the teacher. However, this improved imitation comes with the downside of also inheriting the teacher's biases.

C.3 Detailed Distilled Bag of Words Bias Analysis

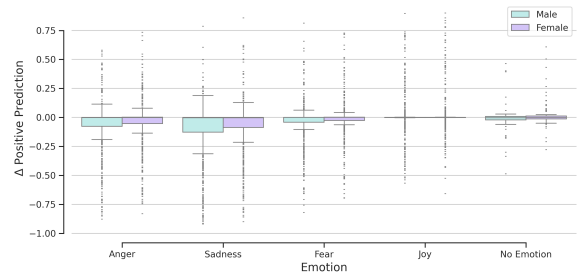


Figure 7: **Bag of Words Gender Difference Between the Distilled and Baseline Model:** This demonstrates the probability of it giving a positive prediction from the distilled version of the model minus the baseline version of the model.

The distilled BoW model stays very close to the baseline: 4.02% of EEC sentences have identical sentiment probabilities and 52.16% change by less than 0.01 in absolute value. Among the remaining sentences, male-gendered variants become more negative on average than female ones (mean shift of -0.0417 vs. -0.0219). Sentences expressing sadness show the strongest changes overall (average shift of -0.0899), with male sentences shifting slightly more than female ones (-0.1016

vs. -0.0783). Race-related differences are comparatively small: European-American, African-American, and race-neutral sentences all exhibit similar sentiment decreases (around -0.09 on average). Other emotions have much smaller shifts (absolute change ≤ 0.023), with anger being the only additional category that shows a noticeable increase in negativity, again somewhat stronger for male terms.

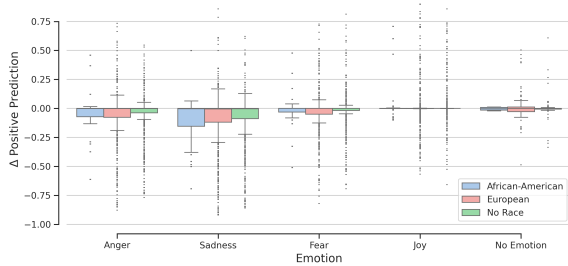


Figure 8: **Bag of Words Race Differences Between the Distilled and Baseline Model:** This demonstrated the difference in the probability of positive prediction from the distilled minus the baseline versions grouped by race.

The paired demographic comparison from the EEC is used to determine whether the distilled BoW model inherits, amplifies, or reduces social bias from the teacher. The resulting gender and race bias measurements are shown in Figures 2 and 1. The distilled BoW model exhibits small but systematic demographic shifts: gender bias becomes slightly more negative for male-associated sentences, consistent with the average probability reductions observed above, while race effects remain minimal, with European-American names showing only a slightly larger sentiment decrease than African-American names. Overall, this suggests that the BoW model inherits some aspects of the teacher’s bias, but the effect is weak compared to the other student architectures.

C.4 Distilled CNN Bias Transfer Graphs

The distilled CNN behaves quite differently from the distilled BoW model. None of the EEC sentences retain identical sentiment probabilities, and only 7.93% change by less than 0.01, indicating substantially higher variance. On average, male-gendered sentences shift only slightly (-0.0056), whereas female-gendered sentences become noticeably more negative (-0.0207), similar in magnitude to the BoW shift for female terms.

Across emotions, sadness again drives the largest negative change (average shift of -0.0529), with male and female variants both becoming more negative. Sentences with no race term show the strongest decrease (-0.0970 on average), while African-American and European-American names exhibit smaller shifts (-0.0375 and -0.0242). Certain configurations (e.g., European male names under fear and joy) show localized increases in positive sentiment, highlighting asymmetric demographic effects. Overall, the distilled CNN model exhibits stronger demographic shifts than BoW (Figures 1 and 2), with more polarized gender effects and larger race-related differences that tend to favor European-American over African-American names. This suggests that the CNN’s greater representational capacity makes it more prone to amplifying the teacher’s latent demographic biases.

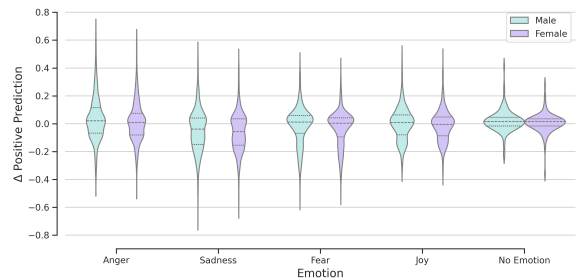


Figure 9: **CNN Gender Difference Between the Distilled and Baseline Model:** This demonstrates the probability of it giving a positive prediction from the distilled version of the model minus the baseline version of the model. Each of the lines inside of the distributions represents a quartile of the data.

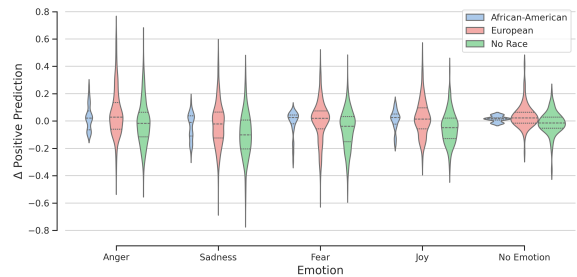


Figure 10: **CNN Race Differences Between the Distilled and Baseline Model:** This demonstrated the difference in the probability of positive prediction from the distilled minus the baseline versions grouped by race. Each of the lines inside of the distributions represents a quartile of the data.

C.5 Distilled DistilBERT Bias Graphs

The distilled DistilBERT model is particularly interesting because the DistilBERT baseline achieves the best IMDB performance, with 96.62% accuracy and an F1 score of 0.9664. In terms of prediction shifts on the EEC, the distilled DistilBERT model falls between the BoW and CNN students: no sentences retain identical sentiment probabilities, 25.35% change by less than 0.01 in absolute value, and 14.55% change by less than 0.001. Overall, the shifts are fairly evenly distributed across demographic groups. Male and female sentences have similar average changes (-0.0922 vs. -0.0951), and African-American and European-American names also show comparable overall shifts (-0.0877 vs. -0.0815).

Unlike the BoW and CNN students, whose largest changes occur for sadness, the distilled DistilBERT model exhibits its largest shift on sentences containing fear, with an average change of -0.2541 —the largest among all models. Sentences with no race term show the strongest decrease (-0.2826), followed by African-American and European-American names (-0.2461 and -0.2337). Overall, the distilled DistilBERT model becomes more confident, especially for sentences with no explicit emotion, where its positive predictions strengthen relative to the baseline.

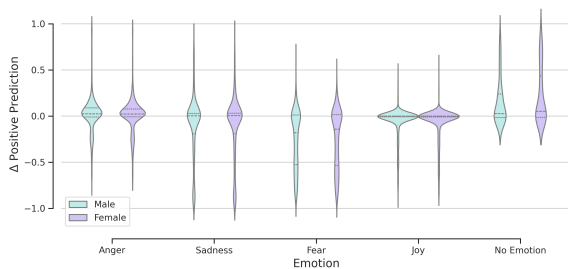


Figure 11: **DistilBERT Gender Difference Between the Distilled and Baseline Model:** This demonstrates the probability of it giving a positive prediction from the distilled version of the model minus the baseline version of the model. Each of the lines inside of the distributions represents a quartile of the data.

Within this general pattern, female sentences without race attributes show some of the largest emotion-specific shifts. For female, race-neutral sentences, the average changes are $+0.2481$ (no emotion), -0.0872 (joy), -0.2836 (fear), and -0.1246 (sadness), with similarly large negative changes for female African-American names under sadness (average -0.1339). Because DistilBERT

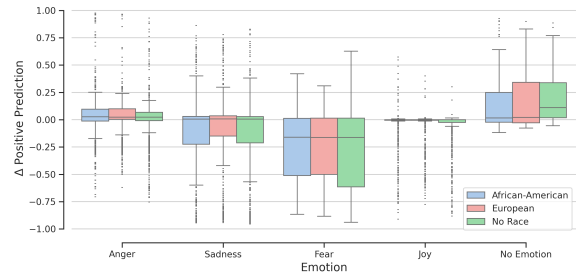


Figure 12: **DistilBERT Race Differences Between the Distilled and Baseline Model:** This demonstrated the difference in the probability of positive prediction from the distilled minus the baseline versions grouped by race.

is the most expressive student model, these results are especially informative: as shown in Figures 1 and 2. Distillation induces large, systematic sentiment shifts with female terms, but it doesn't really inherit any race bias. Both the results for race bias are statistically insignificant. Thus, for DistilBERT, race-related differences appear minimal, and distillation does not seem to transmit the teacher's racial bias.